Diversity Statement

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I promote cultural competency in both teaching and research mentoring settings by **presenting students with different perspectives of what it means to be a computer scientist**. Computer science is a relatively young and burgeoning discipline, with advancements being made by researchers around the world. However, many of the formative contributions to the field were made by white males, a group which still dominates the computer science major in the U.S. [1]. Students from outside this demographic can feel isolated from their peers, or even conclude that they don't belong in the computer science major all together. At the University of Central Florida (UCF), I work to counter this narrative by providing perspectives on computing from a variety of backgrounds, and by conducting research with multi-cultural teams of undergraduate students. While teaching and mentoring, I share my experiences as a first-generation graduate student to show that pursuing a career in research is possible without a background in higher education.

Teaching

UCF is massive school, with more than 68,000 students from diverse ethnic backgrounds. Approximately 40% of UCF students are white, about 30% are Hispanic or Latino, 9% are African American, 8% are Asian, and the remaining 13% identify as multi-racial, or simply as U.S. non-residents [2], [3]. One way that I connect with this diverse group of students is by supplementing my instruction with articles, software, research papers, and other computing resources from an equally diverse set of authors. For instance, when teaching students in my Systems Software course about a cyber attack technique known as stack smashing, I begin by discussing an article written by Venezuelan computer scientist Elias Levy [4]. Sharing this resource with my students not only helps expand their knowledge of computer memory and security, but also helps broaden their awareness of the diverse cultural identities that have progressed the discipline to where it is today.

Research Mentoring

I enjoy seeing my students express interest in research, and am always happy to provide them with opportunities for undergraduate research collaborations, both with myself and other UCF computer science faculty. When conducting research on hardening software build systems, I recruited a team of seven students to assist me with cataloging the software build phases of hundreds of different open-source projects. This team included both male and female students of European, Latin, Vietnamese, and Arabic origin. Over the course of an entire semester we cataloged over 1,000 build phases, and finally produced a dataset serving as the basis for a research paper currently under review at the 49th International Conference on Software Engineering.

Conclusion

I work continuously to improve my intercultural understanding. As a teacher, I provide my students with diverse perspectives on computing in order to help them engage with course material. As a research mentor, I promote cross-cultural connections by forming diverse teams of undergraduate research students, and by sharing my experiences as a first-generation graduate student. And at all times, I listen closely to my students, casting off my own preconceptions about computing in order to better appreciate my students' unique insights, and better answer their questions.

References

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[4] E. Levy, “Smashing The Stack For Fun And Profit.” [Online]. Available: https://insecure.org/stf/smashstack.html